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Chapter 1 – Conditioning and stability

IEEE 754 double-precision format Normalized numbers are written

$$x = \pm(1.b_1 \ldots b_{p-1})_2 \times 2^{(E-1023)},$$

with precision $p = 53$. The significand has 52 bits (plus hidden leading 1), the exponent 11 bits with $1 \le E \le 2046$, and 1 sign bit.

Range The smallest and largest normalized numbers are

$$x_{\min} = 2^{-1022} \approx 2.2{\cdot}10^{-308}, \quad x_{\max} = (2{-}2^{-52}){\times}2^{1023} \approx 1.8{\cdot}10^{308}.$$

Machine precision Gap between 1 and next floating point number,

$$\varepsilon_M = 2^{1-p} = 2^{-52} \approx 2.2 \cdot 10^{-16}.$$

Subnormal numbers These are of the form

$$x = \pm(0.b_1 \ldots b_{p-1})_2 \times 2^{E_{\min}-1023}.$$

These fill the gap below $x_{\min}$, smallest is $2^{-1074} \approx 5 \cdot 10^{-324}$.

Exceptions An exponent $E = 0$ with zero significand encodes ± 0, while $E = 2047$ encodes $\pm\infty$ (if significand=0) or NaN (otherwise).

Rounding (Thm. 1.2) For any x such that $x_{\min} < |x| < x_{\max}$,

$$\hat{x} = x(1+\varepsilon), \quad |\varepsilon| \le \varepsilon_M/2 \quad \text{(nearest)}.$$

Floating point arithmetic (Thm. 1.4) Let x and y be two normalized floating point numbers such that $x_{\min} < |x \star y| < x_{\max}$. Then

$$\widehat{x \star y} = (x \star y)(1+\varepsilon), \quad \star \in \{+,-,\times,/\}, \quad |\varepsilon| \le \varepsilon_M/2 \quad \text{(nearest)}.$$

Relative condition number (Def. 1.9) It is defined by

$$\kappa(x) = \lim_{\varepsilon \to 0} \sup_{\|\delta x\| \le \varepsilon} \left(\frac{\|\delta f\|}{\|f(x)\|} \Big/ \frac{\|\delta x\|}{\|x\|} \right).$$

For differentiable f with Jacobian J, $\kappa(x) = \|J(x)\| \big/ (\|f(x)\|/\|x\|)$.

Forward stability (Def. 1.10) A stable algorithm gives nearly the right answer to nearly the right question, i.e.,

$$\frac{\|\hat{f}(x) - f(\hat{x})\|}{\|f(\hat{x})\|} = \mathcal{O}(\varepsilon_M) \quad \text{with} \quad \frac{\|\hat{x} - x\|}{\|x\|} = \mathcal{O}(\varepsilon_M).$$

Backward stability (Def. 1.11) A backward stable algorithm gives exactly the right answer to nearly the right question, i.e.,

$$\hat{f}(x) = f(\hat{x}) \quad \text{with} \quad \frac{\|\hat{x} - x\|}{\|x\|} = \mathcal{O}(\varepsilon_M).$$

These verify $\|\hat{f}(x) - f(\hat{x})\| / \|f(\hat{x})\| = \mathcal{O}(\kappa(x)\varepsilon_M)$ (Prop. 1.13).

Chapter 2 – Polynomial interpolation

Uniqueness (Thm. 2.1) Given $(n+1)$ distinct $x_0, \ldots, x_n \in \mathbb{R}$ and $(n+1)$ values $y_0, \ldots, y_n \in \mathbb{R}$, there is a unique polynomial p_n of degree at most n such that $p_n(x_i) = y_i$ for all $i = 0, \ldots, n$.

Lagrange polynomials These are defined by

$$\ell_i(x) = \prod_{j \ne i} \frac{x - x_j}{x_i - x_j}$$

and verify $\ell_i(x_j) = \delta_{ij}$. We also define $\Pi_{n+1}(x) = \prod_{i=0}^{n}(x - x_i)$.

Errors for differentiable functions (Thm. 2.2) Assume $f : [a,b] \to \mathbb{R}$ is $\mathcal{C}^{n+1}$. Then for any set of distinct points $x_0, \ldots, x_n$,

$$\|f - p_n(f)\|_\infty \le \frac{\|\Pi_{n+1}\|_\infty}{(n+1)!} \|f^{(n+1)}\|_\infty.$$

Errors for analytic functions (Thm. 2.3) Assume $f : [a,b] \to \mathbb{R}$ is analytic in the disk centered at $(a+b)/2$ with radius $R > 3/2(b-a)$. Then for any set of distinct points $x_0, \ldots, x_n$,

$$\|f - p_n(f)\|_\infty \to 0 \quad \text{as } n \to \infty.$$

Lebesgue cst (Def. 2.10) Defined by $\Lambda_n = \sup_{\|f\|_\infty \le 1} \|p_n(f)\|_\infty$.

Error for L-Lip. functions (Thm. 2.15–2.16) Assume $f : [a,b] \to \mathbb{R}$ is L-Lipschitz. Then for any set of distinct points $x_0, \ldots, x_n$,

$$\|f - p_n(f)\|_\infty \le (1 + \Lambda_n) \frac{3L(b-a)}{2\sqrt{n}}.$$

For Chebyshev points, $\Lambda_n \sim \frac{2}{\pi}\log(n)$, hence $\|f - p_n(f)\|_\infty \to 0$.

Errors for differentiable functions (Cor. 2.18) Assume $f : [a,b] \to \mathbb{R}$ is $\mathcal{C}^k$. Then for Chebyshev points,

$$\|f - p_n(f)\|_\infty \le C_k \frac{\log n}{n^k}.$$

Errors for analytic functions (Thm. 2.19) Assume $f : [a,b] \to \mathbb{R}$ is analytic in the Bernstein ellipse E_ρ. Then for Chebyshev points,

$$\|f - p_n(f)\|_\infty \le C \frac{\sup_{z \in E_\rho} |f(z)|}{\rho^n}.$$

Chapter 3 – Quadrature

Elementary quadrature rule (Def. 3.1) An $(n+1)$-point elementary quadrature rule is a linear form $I_n : \mathcal{C}^0([a,b]) \to \mathbb{R}$ of the form

$$I_n(f) = \sum_{k=0}^{n} w_k f(x_k).$$

Lagrange-based quadrature (Def. 3.3) When $w_k = \int_a^b \ell_k(x)\,dx$. Equidistant/Chebyshev gives Newton–Cotes/Clenshaw–Curtis.

Errors for differentiable functions (Thm. 3.18) For an $(n+1)$-point Lagrange-based rule and $f \in \mathcal{C}^{n+1}([a,b])$,

$$|I(f) - I_n(f)| \le \frac{\|f^{(n+1)}\|_\infty}{(n+1)!} \int_a^b |\Pi_{n+1}(x)|\,dx.$$

Error for analytic functions (Thm. 3.19) Assume $f : [a,b] \to \mathbb{R}$ is analytic in the disk centered at $(a+b)/2$ with radius $R > 3/2(b-a)$. For any sequence of Lagrange-based quadrature rules,

$$|I(f) - I_n(f)| \to 0 \quad \text{as } n \to \infty.$$

Errors for differentiable functions (Thm. 3.25) If f is in $\mathcal{C}^k([a,b])$:

$$|I(f) - I_n(f)| \le \begin{cases} C/n^k, & \text{Clenshaw–Curtis,} \\ C/(2n+1)^k, & \text{Gauss–Legendre.} \end{cases}$$

Errors for analytic functions (Thm. 3.26) If f is analytic in E_ρ:

$$|I(f) - I_n(f)| \le \begin{cases} CM_\rho/\rho^n, & \text{Clenshaw–Curtis,} \\ CM_\rho/\rho^{2n}, & \text{Gauss–Legendre.} \end{cases}$$

Composite quadrature (Def. 3.27) Given a subdivision $a = t_0 < t_1 < \cdots < t_m = b$, a composite quadrature rule is the linear form

$$I_{m,n}(f) = \sum_{j=0}^{m-1} I_n^{[t_j,t_{j+1}]}(f), \quad I_n^{[t_j,t_{j+1}]} \text{ elementary rule.}$$

Trapezoidal rule (Ex. 3.30) The composite trapezoidal rule is

$$I_m(f) = \frac{b-a}{m} \sum_{k=0}^{m-1} \frac{f(x_k) + f(x_{k+1})}{2}, \quad x_k = a + k(b-a)/m.$$

For $f \in \mathcal{C}^2([a,b])$, $I_m(f) = I(f) + \mathcal{O}(m^{-2})$. This is a composite quadrature rule with Newton-Cotes ($n = 1$) on each subinterval.

Chapter 4 – Direct methods for linear systems

Gaussian elimination (Thm. 4.1) For any $A \in \mathbb{R}^{n\times n}$, there exists a permutation matrix P such that

$$PA = LU \qquad (A \in \mathbb{R}^{n\times n}),$$

where L is a unit lower triangular matrix ($\ell_{ii} = 1$) and U is upper triangular. If A is nonsingular, U is nonsingular and the factorization can be used to solve $Ax = b$. Computational cost is $\mathcal{O}(2n^3/3)$.

Condition number κ (Thm. 4.3) Let $A \in \mathbb{R}^{n\times n}$ be nonsingular, and consider the systems $Ax = b$ and $\hat{A}\hat{x} = \hat{b}$ with

$$\|\hat{A} - A\| \le \varepsilon_A \|A\|, \qquad \|\hat{b} - b\| \le \varepsilon_b \|b\|, \qquad (\text{e.g., } \varepsilon_A, \varepsilon_b \le \varepsilon_M).$$

If $\varepsilon_A \kappa(A) < 1$, then

$$\frac{\|\hat{x} - x\|}{\|x\|} \le \frac{\kappa(A)}{1 - \varepsilon_A \kappa(A)} (\varepsilon_A + \varepsilon_b),$$

where $\kappa(A) = \|A\|\,\|A^{-1}\|$ is the condition number of A.

Properties of κ (Prop. 4.4) Let $A \in \mathbb{R}^{n\times n}$ be nonsingular. Then

(a) $\kappa(A) \ge 1$;

(b) $\kappa(\alpha A) = \kappa(A)$ for any $\alpha \neq 0$;

(c) $\kappa(A) = \dfrac{\max_{\|y\|=1} \|Ay\|}{\min_{\|z\|=1} \|Az\|}$.

Property (c) allows extending the definition to rectangular matrices.

Stability of Gaussian elimination (Thm. 4.5) Let $A \in \mathbb{R}^{n\times n}$ be nonsingular, and let $\hat{A} = \hat{L}\hat{U}$ be computed by Gaussian elimination with pivoting ($|\hat{\ell}_{ij}| \le 1$). Then, up to $\mathcal{O}(\varepsilon_M^2)$,

$$|\hat{a}_{ij} - a_{ij}| \le 2a \min(i-1, j)\varepsilon_M, \quad a = \max_{i,j,k} |\hat{a}_{ij}^{(k)}|.$$

Cholesky factorization (Thm. 4.6) Let $A \in \mathbb{R}^{n\times n}$ be symmetric positive definite (SPD). Gaussian elimination can be performed without pivoting and yields the Cholesky factorization

$$A = LL^T \qquad (\text{SPD } A \in \mathbb{R}^{n\times n}),$$

where L is lower triangular. Computational cost is $\mathcal{O}(n^3/3)$.

Stability of Cholesky factorization (Thm. 4.7) Let $A \in \mathbb{R}^{n\times n}$ be SPD, and let $\hat{A} = \hat{L}\hat{L}^T$. Then, up to $\mathcal{O}(\varepsilon_M^2)$,

$$|\hat{a}_{ij} - a_{ij}| \le a_0 \min(i, j)\varepsilon_M, \quad a_0 = \max_{i,j} |a_{ij}| = \max_i |a_{ii}|.$$

Normal equations (Thm. 4.8) $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $m > n$. Then

$$\|Ax - b\|_2 \to \min \quad \Longleftrightarrow \quad A^TAx = A^Tb \quad (\text{normal equations}).$$

The normal equations always have at least one solution. Moreover, A^TA is symmetric and non-negative since $x^TA^TAx = \|Ax\|_2^2 \ge 0$.

QR Factorization For any $A \in \mathbb{R}^{m\times n}$ with $m \ge n$, there exists an orthogonal matrix Q and an upper triangular matrix R such that

$$A = QR \qquad (A \in \mathbb{R}^{m\times n}).$$

For least squares problems, only the first n columns of Q are needed. Computational cost is $\mathcal{O}(2mn^2 - 2n^3/3)$.

Solving the normal equations

1. Cholesky factorization: Compute $A^TA = LL^T$ and solve
$$x = L^{-T}L^{-1}A^Tb.$$
This is faster, $\mathcal{O}(mn^2 + n^3/3)$ flops, but squares the condition number $\kappa(A^TA) = \kappa(A)^2$, hence less stable.

2. QR factorization: Compute $A = QR$ and solve
$$x = R^{-1}Q^Tb.$$
This is slower, $\mathcal{O}(2mn^2 - 2n^3/3)$ flops, but more stable.

Chapter 5 – Iterative methods for linear systems

Direct vs iterative For sparse PDE systems, direct methods are efficient in 1D and 2D. In 3D, however, fill-in makes LU/Cholesky factorization costly in both memory and computation, so iterative methods with preconditioning are generally preferred.

Principle Consider $Ax = b$ in $\mathbb{R}^n$. Iterative methods generate a sequence $(x^k)_{k\in\mathbb{N}}$ converging to the solution x^∞ of $Ax^\infty = b$, typically via a fixed-point scheme $x^{k+1} = f_k(x^k)$.

- Stationary: x^{k+1} is a fixed combination of x^k and b, e.g.,
$$Mx^{k+1} = Nx^k + b \qquad (A = M - N \text{ with cheap } M^{-1}).$$
- Non-stationary: the update $f_{k+1} \neq f_k$ varies at each iteration.

Convergence (Cor. 5.11) The iterative sequence

$$x^{k+1} = Px^k + c$$

converges for any $x^0 \in \mathbb{R}^n$ if and only if $\rho(P) < 1$. In that case

$$\lim_{k\to\infty} x^k = (I - P)^{-1}c.$$

Remark Substituting $P = M^{-1}N$ and $c = M^{-1}b$ gives

$$x^\infty = (I - M^{-1}N)^{-1}M^{-1}b.$$

For $A = M - N$, we have $M(I - M^{-1}N) = A$, hence $x^\infty = A^{-1}b$.

Gershgorin disks (Thm. 5.14) Let $A \in \mathbb{R}^{n\times n}$, and λ an eigenvalue of A. Then there exists $1 \le i \le n$ such that $|\lambda - a_{ii}| \le \sum_{j\neq i} |a_{ij}|$.

Jacobi Choose $M = D$ (diagonal of A) and $N = D - A$:

$$x^{k+1} = (I - D^{-1}A)x^k + D^{-1}b.$$

Convergence: $\rho(I - D^{-1}A) < 1$ (Cor. 5.25); true if A is SRDD.

Gauss–Seidel Choose $M = L + D$ (lower triangular part plus diagonal) and $N = L + D - A = -U$ (strict upper triangular part):

$$x^{k+1} = -(L + D)^{-1}Ux^k + (L + D)^{-1}b.$$

Convergence: $\rho((L + D)^{-1}U) < 1$; true if A is SRDD or SPD.

Spectral radius For $A \in \mathbb{R}^{n\times n}$, $\rho(A) = \max\{|\lambda| : \lambda \in \mathrm{Sp}(A)\}$.

- $\lim_{k\to\infty} \|A^k\|^{1/k} = \rho(A)$;
- $\rho(A) = \rho(S^{-1}AS)$ for any nonsingular S.

Matrix norms For $A \in \mathbb{R}^{m\times n}$, the induced (operator) norm is

$$\|A\|_p = \sup_{\|x\|_p \le 1} \|Ax\|_p.$$

In particular, we have the simpler formulas:

- $\|A\|_1 = \max_j \sum_i |a_{ij}|$ (maximum column sum);
- $\|A\|_\infty = \max_i \sum_j |a_{ij}|$ (maximum row sum);
- $\|A\|_2 = \sqrt{\rho(A^TA)}$ (spectral norm).

An example of a non-induced norm is the Frobenius norm

$$\|A\|_F = \left(\sum_{i,j} |a_{ij}|^2\right)^{1/2} = \sqrt{\mathrm{tr}(A^TA)}.$$

Properties

- $\|AB\| \le \|A\|\|B\|$ for induced and Frobenius norms.
- $\|A\|_2 = \rho(A)$ for symmetric A.
- $\|QA\|_2 = \|A\|_2$ and $\|QA\|_F = \|A\|_F$ for orthogonal Q.

Chapter 6 – Solving nonlinear equations

Setup in $\mathbb{R}$ Let $f : \mathbb{R} \to \mathbb{R}$ be $\mathcal{C}^1$. We want to find a numerical solution to $f(x) = 0$. The true solution will be denoted by x_*.

Newton's method Given $x_0 \in \mathbb{R}$, the iterates are defined by

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad n \geq 0.$$

Secant method Given $x_0, x_1 \in \mathbb{R}$, the iterates are defined by

$$x_{n+1} = x_n - \frac{f(x_n)}{g(x_n)}, \quad g(x_n) = \frac{f(x_n) - f(x_{n-1})}{x_n - x_{n-1}}, \quad n \geq 1.$$

Convergence via contraction (Thm. 6.3) For $x_{n+1} = g(x_n)$ with fixed point $g(x_*) = x_*$, convergence holds where $|g'(x)| < 1$.

Convergence order (Def. 6.4) It is said to be of order $m \geq 1$ if

$$|x_{n+1} - x_*| = \mathcal{O}(|x_n - x_*|^m).$$

Newton convergence (Thm. 6.5) Let $f : [a, b] \to \mathbb{R}$ be $\mathcal{C}^2$ with a simple root x_* ($f(x_*) = 0$, $f'(x_*) \neq 0$). Then there exists $r > 0$ such that, for any $x_0 \in [x_* - r, x_* + r]$, Newton's iterates are well defined, stay in the interval, and converge quadratically to x_*.

Secant convergence (Thm. 6.7) Under the same assumptions, secant iterates converge superlinearly to x_*.

Stopping criteria The error $|x_{n+1} - x_*|$ is usually unavailable; practical criteria are the residual $|f(x_{n+1})|$ or the increment $|x_{n+1} - x_n|$.

Conditioning If f depends on data d, e.g., $d = (A, b)$ for linear systems, then the condition numbers can be approximated by

$$\kappa(d) \approx \frac{|d|}{|x_*||f'(x_*)|}, \quad \kappa_{\text{abs}}(d) \approx \frac{1}{|f'(x_*)|}.$$

This yields

$$\frac{|\hat{x} - x_*|}{|x_*|} \lesssim \frac{|f(\hat{x})|}{|x_*||f'(x_*)|}.$$

If well-conditioned, the residual $|f(\hat{x})|$ thus estimates the error well.

Setup in $\mathbb{R}^n$ Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be $\mathcal{C}^1$. We want to find a numerical solution to $F(\boldsymbol{x}) = 0$. The true solution will be denoted by $\boldsymbol{x}_*$.

Newton's method Given $\boldsymbol{x}_0 \in \mathbb{R}^n$, the iterates are defined by

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n - J_F(\boldsymbol{x}_n)^{-1} F(\boldsymbol{x}_n), \quad n \geq 0.$$

Convergence order It is said to be of order $m \geq 1$ if

$$\|\boldsymbol{x}_{n+1} - \boldsymbol{x}_*\| = \mathcal{O}(\|\boldsymbol{x}_n - \boldsymbol{x}_*\|^m).$$

Newton convergence (Thm. 6.10) Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be $\mathcal{C}^1$ on a convex open set D containing $\boldsymbol{x}_*$. Assume $J_F(\boldsymbol{x}_*)^{-1}$ exists with $\|J_F(\boldsymbol{x}_*)^{-1}\| \leq C$ and J_F is L-Lipschitz in $B(\boldsymbol{x}_*; R)$. Then there exists $r > 0$ such that, for any $\boldsymbol{x}_0 \in B(\boldsymbol{x}_*; r)$, Newton's iterates are well defined, stay in the ball, and converge quadratically to $\boldsymbol{x}_*$.

Modified Newton's method To speed up Newton's method:

- freeze J_F for several iterations and reuse its LU factorization;
- solve inexactly with the exact J_F (e.g., a few Jacobi/SOR iterations yield Newton–Jacobi/Newton–SOR);
- solve exactly with an inexact J_F (e.g., finite differences).

Quasi-Newton methods Quasi-Newton methods approximate the Jacobian via updates $G_n \approx J_F(\boldsymbol{x}_n)$. Let $\delta F_n = F(\boldsymbol{x}_n) - F(\boldsymbol{x}_{n-1})$ and $\delta \boldsymbol{x}_n = \boldsymbol{x}_n - \boldsymbol{x}_{n-1}$. For instance, Broyden's method reads

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n - G_n^{-1} F(\boldsymbol{x}_n), \quad G_n = G_{n-1} + \frac{\delta F_n - G_{n-1}\delta \boldsymbol{x}_n}{\|\delta \boldsymbol{x}_n\|^2} \delta \boldsymbol{x}_n^T.$$

Often initialized with $G_0 = J_F(\boldsymbol{x}_0)$. Superlinear convergence under the assumptions of Thm. 6.10 with $G_0 \approx J_F(\boldsymbol{x}_*)$ (Prop. 6.13).

Chapter 7 – ODEs I

Setup Consider, on $[t_0, t_0 + T]$, $y' = f(t, y)$ with $y(t_0) = y_0$.

Cauchy–Lipschitz (Thm. 7.5) If f is continuous and locally Lipschitz in y, there is a unique solution y, defined on all of $[t_0, t_0 + T]$ if f is globally Lipschitz in y. Moreover, $f \in \mathcal{C}^k \Rightarrow y \in \mathcal{C}^{k+1}$.

Time discretization Time is discretized by $N + 1$ points

$$t_0 < t_1 < \ldots < t_N = t_0 + T,$$

with $\Delta t_n = t_{n+1} - t_n$, $\Delta t = \max \Delta t_n$. For $t_0 = 0$ and $\Delta t_n = \Delta t$:

$$t_0 = 0 < \ldots < t_n = n\Delta t < \ldots < t_N = T, \quad \Delta t = T/N.$$

General one-step methods These are of the form

$$y_{n+1} = y_n + \Delta t_n \, \phi(t_n, y_n; \Delta t_n).$$

For instance (see Ex. 8.2 when ϕ is defined implicitly):

$$y_{n+1} = y_n + \Delta t_n f(t_n, y_n), \qquad \text{(forward Euler)},$$
$$y_{n+1} = y_n + \Delta t_n f(t_{n+1}, y_{n+1}), \qquad \text{(backward Euler)}.$$

Errors & convergence (Def. 7.7) The *global* error is

$$e_n = y(t_n) - y_n, \qquad n = 0, \ldots, N.$$

The method is convergent if

$$\lim_{\Delta t \to 0} \left(\max_{n=0,\ldots,N} |e_n| \right) = 0.$$

The *local truncation* error or simply *local* error is defined by

$$\varepsilon_n = y(t_{n+1}) - y(t_n) - \Delta t_n \, \phi(t_n, y(t_n); \Delta t_n), \quad n = 0, \ldots, N-1.$$

The *consistency* error is defined by

$$E = \sum_{n=0}^{N-1} |\varepsilon_n|.$$

Consistency (Def. 7.9) The method is consistent if

$$|\varepsilon_n| \leq \Delta t_n \Psi(\Delta t), \qquad \Psi(\Delta t) \to 0 \quad \text{as} \quad \Delta t \to 0.$$

Convergence order (Def. 7.10) The scheme is of order $p \geq 1$ if

$$|\varepsilon_n| = \mathcal{O}(\Delta t_n^{p+1}), \qquad n = 0, \ldots, N-1.$$

Then $E \leq C \Delta t^p \sum_n \Delta t_n = CT \Delta t^p$ (hence Δt_n in Def. 7.9).

Stability to perturbations (Def. 7.12) If there exists $C > 0$, independent of Δt, such that for any perturbed iteration

$$\xi_{n+1} = \xi_n + \Delta t_n \, \phi(t_n, \xi_n; \Delta t_n) + \eta_n,$$

it holds that

$$\max_{n=0,\ldots,N} |y_n - \xi_n| \leq C \left(|y_0 - \xi_0| + \sum_{n=0}^{N-1} |\eta_n| \right).$$

Also called zero-stability (for multistep: bounded $\{y_n\}$ for $f = 0$).

Forward Euler We assume that f is L-Lipschitz.

1. Consistent, of order 1 if y is $\mathcal{C}^2$ (Prop. 7.11).
2. Stable with $C = e^{LT}$ (Prop. 7.14).
3. Convergent (Thm. 7.15), order 1 if y is $\mathcal{C}^2$ (Thm. 7.15).

Backward Euler We assume that f is L-Lipschitz.

1. Consistent if $L\Delta t < 1$, of order 1 if y is $\mathcal{C}^2$ (Prop. 7.28).
2. Stable with
$$C = \frac{e^{LT/(1-L\Delta t)}}{1 - L\Delta t} \quad \text{(Prop. 7.27)}.$$
3. Convergent (Thm. 7.31), order 1 if y is $\mathcal{C}^2$ (Thm. 7.32).

Absolute stability Abs. stable for $\lambda \Delta t$ if bounded $\{y_n\}$ for $f = \lambda y$.

Chapter 8 – ODEs II

Consistency/Order/Stability Consistency (Def. 8.3), convergence order (Def. 8.4), and stability (Def. 8.8) are defined as previously.

RK methods Of the form $y_{n+1} = y_n + \Delta t_n\, \phi(t_n, y_n; \Delta t_n)$.

- RK1 (Euler method):

$$y_{n+1} = y_n + \Delta t_n f(t_n, y_n).$$

- RK2 (midpoint method):

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \tfrac{\Delta t_n}{2}, y_n + \tfrac{\Delta t_n}{2} k_1\right), \\ y_{n+1} &= y_n + \Delta t_n k_2. \end{aligned}$$

- RK4:

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f\left(t_n + \tfrac{\Delta t_n}{2}, y_n + \tfrac{\Delta t_n}{2} k_1\right), \\ k_3 &= f\left(t_n + \tfrac{\Delta t_n}{2}, y_n + \tfrac{\Delta t_n}{2} k_2\right), \\ k_4 &= f(t_n + \Delta t_n, y_n + \Delta t_n k_3), \\ y_{n+1} &= y_n + \tfrac{\Delta t_n}{6}\left(k_1 + 2k_2 + 2k_3 + k_4\right). \end{aligned}$$

Convergence (Thm. 8.10) If the one-step method is stable with respect to perturbations and consistent, then it is convergent. If it is C-stable and of order p, and if f is of class $\mathcal{C}^p$, then

$$\max_{0 \le n \le N} |e_n| \le C\left(|e_0| + K \Delta t^p\right) \quad \text{with } K > 0 \text{ indep. of } \Delta t.$$

Stability under perturbations (Thm. 8.11) Assume that ϕ is uniformly L-Lipschitz in y with respect to t and Δt. Then the one-step method is stable under perturbations with $C = \exp(LT)$.

One-step method in vector form For the test equation $y' = \lambda y$:

$$y_{n+1} = r(\lambda \Delta t_n) y_n.$$

For instance, for RK1 and RK2 (Ex. 8.23):

$$r(z) = 1 + z \quad \text{(RK 1)}, \qquad r(z) = 1 + z + \frac{z^2}{2} \quad \text{(RK 2)}.$$

If a Runge–Kutta method is of order p then (Lem. 8.24)

$$r(z) = \exp(z) + C z^{p+1} + O(z^{p+2}).$$

Absolute stability domain (Def. 8.25) It is the set

$$D = \{z \in \overline{\mathbb{C}} \,:\, |r(z)| \le 1\},$$

where $\overline{\mathbb{C}}$ denotes the extended complex plane, i.e., the complex plane with a point at infinity ∞. The interior is denoted by $\mathring{D}$.

$\mathcal{A}$-stability (Def. 8.26) An RK method is said to be $\mathcal{A}$-stable if

$$\{z \in \overline{\mathbb{C}} \,:\, \Re(z) \le 0 \text{ or } z = \infty\} \subset D.$$

Using the maximum principle on $r(z)$, one can show that a method is $\mathcal{A}$-stable if and only if (Rem. 8.27)

$$\{z \in \mathbb{C} \,:\, \Re(z) < 0\} \subset \mathring{D}.$$

In practice, it suffices to check that r has no poles in $\{\Re(z) < 0\}$ and that $|r(iy)| \le 1$ for all $y \in \mathbb{R}$.

Explicit Runge–Kutta methods (Rem. 8.28) Note that an explicit Runge–Kutta method is never $\mathcal{A}$-stable, since $r(z)$ is a polynomial and its absolute stability domain is necessarily bounded.

Order and stability For explicit RK methods up to order four, the stability region grows with order, unlike multistep explicit Adams–Bashforth methods, whose stability region shrinks with order.

Chapter 9 – PDEs

Heat equation Consider the heat equation:

$$\begin{cases} u_t - \nu u_{xx} = 0, & t \ge 0,\ x \in [0,1], \\ u(0,x) = u_0(x), & x \in [0,1]. \end{cases}$$

To have a unique solution, we must provide boundary conditions. Common choices are Dirichlet (specify u) or Neumann (specify u_x).

Discretization We consider the interval $[0,1]$ and define

$$\Delta x = 1/(I+1), \quad x_0 = 0,\ \ldots,\ x_i = i\Delta x,\ \ldots,\ x_{I+1} = 1.$$

For the time interval $[0,T]$,

$$\Delta t = T/(N+1), \quad t_0 = 0,\ \ldots,\ t_n = n\Delta t,\ \ldots,\ t_{N+1} = T.$$

Finite difference method Approximate $u(t_n, x_i)$ by U_i^n using an explicit Euler in time and second-order centered differences in space:

$$\begin{cases} U_i^0 = u_0(x_i), \\ \dfrac{U_i^{n+1} - U_i^n}{\Delta t} - \nu \dfrac{U_{i-1}^n - 2U_i^n + U_{i+1}^n}{\Delta x^2} = 0. \end{cases}$$

Consistency (Def. 9.13) Let ε_i^n denote the local truncation error obtained by inserting the exact $u_i^n := u(t_n, x_i)$ into the scheme:

$$\varepsilon_i^n := \frac{u_i^{n+1} - u_i^n}{\Delta t} - \nu \frac{u_{i-1}^n - 2u_i^n + u_{i+1}^n}{\Delta x^2}.$$

The scheme is said to be consistent if

$$\varepsilon_i^n \to 0 \quad \text{as } \Delta t \to 0 \text{ and } \Delta x \to 0.$$

Order q in time and order p in space if $\varepsilon_i^n = O(\Delta t^q) + O(\Delta x^p)$.

Stability (Def. 9.17) Stable in a norm $\|\cdot\|$ if, for all $T > 0$, there exists a constant $K_T > 0$, independent of Δt and Δx, such that

$$\|U^n\| \le K_T \|U^0\|, \quad 0 \le n\Delta t \le T,$$

for any initial data U^0. If this holds only under conditions on Δt and Δx, it is said to be conditionally stable.

Stability with matrix norm (Prop. 9.18) Write the discretization scheme as $U^{n+1} = MU^n$. Stability is then equivalent to

$$\|M^n\| \le K, \quad n = 0, \ldots, N+1,$$

independently of Δx and Δt (possibly under some conditions). Since $\|M^n\| \le \|M\|^n$, a sufficient condition is $\|M\| \le 1$.

Convergence (Thm. 9.25) Let $u(t,x)$ be a sufficiently smooth solution of the heat equation with given initial and boundary conditions. Consider a linear two-level scheme (such as above) that is consistent and stable in a norm $\|\cdot\|$. Then the scheme is convergent:

$$\forall T > 0, \qquad \lim_{\Delta t \to 0,\, \Delta x \to 0} \sup_{t_n \le T} \|e^n\| = 0,$$

where $e_j^n = U_j^n - u_j^n$ is the numerical error. Moreover, if the scheme is of order q in time and p in space, then

$$\sup_{t_n \le T} \|e^n\| \le C_T\left(\Delta t^q + \Delta x^p\right).$$

Proving ℓ^2-stability For periodic boundary conditions, expand

$$U^n(x) = \sum_{k \in \mathbb{Z}} \hat{U}^n(k) e^{2i\pi k x},$$

and substitute into the scheme. Since $U^n(x \pm \Delta x)$ has coefficients $\hat{U}^n(k) e^{\pm 2i\pi k \Delta x}$, each mode evolves independently:

$$\hat{U}^{n+1}(k) = g(k)\hat{U}^n(k), \quad g(k) = 1 - 4\sigma \sin^2(\pi k \Delta x), \quad \sigma = \frac{\nu \Delta t}{\Delta x^2}.$$

By Parseval, ℓ^2-stability holds iff $|g(k)| \le 1$ for all k, i.e.,

$$\Delta t \le \frac{\Delta x^2}{2\nu}.$$

Thus the explicit scheme is conditionally ℓ^2-stable.